

# Cadence: learning through local repair

Overlap repair, equilibrium detuning, and a continuing actor

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## Abstract

Cadence studies learning through bounded observer-like patches that retain state, exchange readings at interfaces and repair inconsistent predictions. Its organizing operations are overlap repair, local equilibrium detuning and functional self-readback. The strongest controlled result combines a body relation acquired from real position/action experience with joint future-state and action repair. Across three learned models, feedback planning reduces disturbed executed cost to 0.02085, compared with 0.05594 for myopic control and 1.23467 for an initial open-loop plan. A fixed linear-Gaussian past model admits an exact compressed boundary: one retained observation and a Gaussian prefix reproduce longer-window estimates and actions, reducing the measured observer payload from 272 to 208 bytes. A scalar Lean proof states the corresponding quadratic elimination identity and its fixed-model hypotheses. These results establish useful learned-model action and conditional memory compression. General lifelong memory, reliable long autonomous continuation and composition remain unestablished. The architecture therefore separates measured observations, learned relationships, retained context and private imagined futures, with explicit tests of whether each changes subsequent action.

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# 1 A continuing actor built from local repair

The design question is the smallest persistent state and update rule that preserve useful experience. A capable continuing learner must acquire a body model, retain context and important relationships, imagine consequences, act, and correct itself from what actually happens. Cadence investigates whether these functions can share bounded patch operations instead of requiring a separate mechanism for each task. The individual is the continuing organization of activity, learned relationships, interfaces and body coupling; one scalar energy or a stable output is not a complete description of it.

Observer-Patch Holography (OPH) supplies the organizing constraint: bounded, self-reading patches exchange declared public readings and repair disagreement while protecting observations [25, 26]. Overlap consensus concerns compatible interface values, not identical internal states. Conditional normal-form results distinguish a consistent answer from an answer independent of the initial interior. Neither property establishes that the input evidence is true or that the learned model is useful. Section A.2 gives the exact neural specialization.

Three operations structure the proposal. *Repair* settles local state under held boundaries. *Detuning* changes relationships using local contrasts between equilibria perturbed by real teaching evidence. *Self-readback* makes proposed actions and predicted consequences available to the same continuing actor, whose next observation tests those predictions. Physical readback supplies new evidence; an imagined outcome does not. A preference can guide future action without rewriting the observed past.

These operations have established antecedents. Hopfield networks compute by collective relaxation [10]; equilibrium propagation derives local credit under symmetry and stable-branch assumptions [16, 30]; temporal predictive coding uses transition and readout residuals [18]. Integrated neural architectures and learned body models motivate the whole-actor goal [6, 7]. Cadence’s contribution is an explicit patch/evidence contract, mathematical boundaries on retained state, and controlled tests of which combinations actually support useful action. No superiority over conventional filtering, control or regression follows from sharing this description.

## 1.1 Scope of the evidence

The base library exposes a nonlinear temporal learner, explicit response constraints, private continuations and a fixed linear-body actor. These share residual-repair ideas but retain distinct state spaces and hypotheses. The nonlinear learner accepts time-varying observations and carries only target-free activity after detuning; the actor holds a fixed body model while separating actual records from hypothetical plans. An audio application must learn or supply its instrument relation. Sources, frozen protocols, controls and receipts identify which implementation supports each result.

The main evidence is presented in Section 3. Appendix A preserves the unique recurrence, record-interference and wiring theorems with their hypotheses. Appendix E retains the source-bound receipt map, including the application evidence that motivated these tests. The public proof package contains 169 audited statements, including the causal-path, quadratic-boundary and protected-path results. Numerical verification is separate from a formal refinement proof of the implementation.

For Amen composition, the actor must use heard context, retained musical intention, imagined alternatives and action consequences together. One-step prediction or a nonconstant generated sequence is insufficient. The relevant criterion is a learned organization that changes its actual continuation correctly when memory, intention or readback is intervened upon, while preserving unrelated learned material.

# 2 Shared patch operations and persistent state

## 2.1 Local state, learned relationships and ports

The library’s **PatchNet** interface wraps the existing nonlinear **Brain** and reciprocal **Learner** [20]. It owns a continuing free neural state, learned synaptic efficacies and biases, and the learner’s optimizer state. Input and output ports name the coordinates through which observations and predictions cross its boundary. A caller supplies their meaning, the initial wiring and the teaching objective. Internal functional specialization is a learning question; declaring a region to be a workspace does not establish that it performs a useful workspace function.

For the fast state, the concrete local constraint is

$$v_i = \sum_j W_{ij} \text{act}(v_j) + c_i,$$

with the nonlinear activation and repair step specified in Section A. A patch reads its neighbours’ published values and its own boundary drive, then updates its discrepancy. Effective weights, rather than neuron labels or equal neighbouring values, define the relationship being repaired. The software schedules these local dependencies centrally. An energy interpretation requires the effective reciprocal interactions and the activation assumptions of the learning derivation; arbitrary directed connectomes do not acquire that interpretation by sharing the interface.

The retained computational state has the form

$$S_t = (v_t, \theta_t, o_t, P_t),$$

where  $v_t$  is free activity,  $\theta_t$  the learned relationships,  $o_t$  the optimizer state and  $P_t$  a bounded optional record of delivered source identifiers. Fixed interface and solver configuration accompany this state. A complete reproducible system also accounts for any external random generator, body state and scheduler that affect subsequent observations. The optional identifier record provides duplicate-delivery bookkeeping; it is not an episodic content store.

The core does not attach the sparse records cortex used by the earlier applications. It also does not binarize synapses or add a separate importance classifier. Continuous learned weights can express intermediate relationships that a fixed binary-contact prior would eliminate. This is a smaller integration of existing Cadence mechanisms, with a specific empirical question: which useful memories can its activity and parameters actually acquire, preserve and revise?

| Operation           | State and evidence contract                                                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>create</b>       | Supplies a reciprocal initial graph and named input/output ports; these are inherited structure, not learned roles.                                                       |
| <b>settle, read</b> | Advances and reads target-free activity without changing learned parameters. Inputs are soft drives and receive recurrent feedback.                                       |
| <b>observe</b>      | Uses externally observed target coordinates and supplied gains; updates parameters only after all required phases converge. A rejected attempt can advance free activity. |
| <b>imagine</b>      | Evolves a sequential private branch; neither live continuation state nor witnessed evidence is changed.                                                                   |
| <b>reset</b>        | Clears fast activity while retaining parameters, optimizer state and delivery identifiers.                                                                                |
| <b>save, load</b>   | Preserves continuation state; exact numerical continuation requires the same backend and precision.                                                                       |

Table 1: The neural core’s public interaction contract. These operations expose computation and evidence boundaries; they do not guarantee learned competence.

## 2.2 An explicit temporal overlap

Carrying an initial state cannot preserve its information when complete settlement has one solution independent of that state. The optional temporal boundary makes the previous free activity  $\bar{s}$  an argument of the constraints themselves. With a supplied mask  $a_i \in \{0, 1\}$  and strength  $\kappa \geq 0$ , its energy term is

$$E_{\theta}^{\bar{s}}(s) = E_{\theta}(s) + \frac{\kappa}{2} \sum_i a_i (s_i - \bar{s}_i)^2.$$

The corresponding local recurrent equation adds  $\kappa a_i (\bar{s}_i - s_i)$  to the drive. The same  $\bar{s}$  is held throughout the free, positive and negative phases of one observed event. It is not advanced at each numerical iteration and is independent of the target’s perturbation strength and evidence gain. The next event uses the preceding free

activity; a fresh or reset stream uses a zero boundary. An imagined branch advances only its private previous state.

The default strength is zero, and the default eligible coordinates are the inherited hidden ports. Positive strength is therefore a supplied persistence assumption. It introduces no additional fact store and reuses existing activity, but it changes the model’s equations and must be counted as a structural choice. The unmodified recurrence’s contraction certificate does not automatically certify this changed solver. In a single-minimum landscape, repeated temporal overlap can merely slow forgetting. It does not discover a protected subspace, choose important events or guarantee indefinite retention. The gradient estimate below is conditional on the held boundary; it does not differentiate through preceding events or solve long-delay temporal credit.

An exact scalar example distinguishes persistence from protection. For  $a > 0$  and  $\kappa \geq 0$ , the energy  $az^2/2 + \kappa(z - c)^2/2$  has unique minimizer  $qc$ , where  $q = \kappa/(a + \kappa) \in [0, 1)$ . Reusing that minimizer as the next boundary gives  $q^n c$ , tending to zero. These statements are machine-checked<sup>1</sup>. They describe a convex scalar system; they do not prove that every nonlinear neural basin washes out.

**A readable trace is a different boundary.** The library also provides **Trace**, used by earlier controllers but not automatically instantiated by **PatchNet**. With movement weighting disabled, it records selected free activity  $h_t$  as

$$z_{t+1} = \rho z_t + (1 - \rho)h_t, \quad 0 \leq \rho < 1,$$

and makes  $z_t$  readable at declared context ports during the next event. The trace is held fixed across all numerical iterations and nudged phases, then advanced once from the free state. This supplies an explicit fading observation history; carrying a numerical initial state or tethering the immediately preceding activity implements a different operation.

For fixed  $z_t, h_t$ , the displayed update is the unique minimizer in  $u$  of

$$\frac{\rho}{2}(u - z_t)^2 + \frac{1 - \rho}{2}(u - h_t)^2.$$

Six machine-checked lemmas establish this scalar repair, interval preservation, decay of an initial difference as  $\rho^n$  under the same source sequence, and decay to zero under zero source<sup>2</sup>. Here the timescale and source are supplied. If the recurrent source itself depends on the trace, the same-source premise need not hold. These lemmas certify neither a joint energy for the full recurrent system nor its stability, learned importance or permanent memory [R80].

## 2.3 Readback and detuning as the learning operation

A witnessed target  $y$  and an observed-coordinate mask  $m$  define a port loss, for example

$$L(s; y, m) = \frac{1}{2} \sum_k m_k (s_k - y_k)^2.$$

Unobserved coordinates do not supply targets. At fixed parameters and fixed temporal boundary, the default centred learner obtains a free state and states under positive and negative perturbations  $E_\theta^\pm \pm \beta L$ . It then estimates a parameter derivative from the difference of the corresponding local energy derivatives:

$$g_e(\beta) = \frac{\partial_{\theta_e} E_\theta(s^+) - \partial_{\theta_e} E_\theta(s^-)}{2\beta}.$$

For the implemented pair interaction, this is an endpoint activity contrast. The current optimizer and parameter constraints determine the ensuing update. The gradient interpretation is imported from equilibrium propagation under its smooth stable-branch, symmetry and convergence assumptions [16, 30]; endpoint locality alone proves none of those assumptions. Finite nudges are biased, branches can differ in a multibasin system, and free-state convergence does not imply convergence of both nudged phases.

The revised interface checks the equation residual of every required phase and refuses a parameter update when a phase reaches its work cap without meeting tolerance. It reports convergence and work rather than

<sup>1</sup>Lean: `repair_unique`, `iterate_closed_form`, `iterate_tends_to_zero`.

<sup>2</sup>Lean: `unique_minimizer`, `repair_mem_interval`, `same_source_history_difference`, `zero_source_tends_to_zero`.

treating small successive movement as sufficient evidence of settlement. This is a numerical acceptance contract. Outside the contractive subclass it is neither a bound on distance to a unique equilibrium nor a proof of local stability or correct task predictions.

The distinction between activity repair and learning is essential. A free equilibrium answers the current constraints. A witnessed teaching perturbation changes persistent relationships only through the learning operation. An unobserved read does not supply a fabricated target. The caller’s loss, target and evidence selection therefore determine what counts as a useful correction; a harmful-outcome nudge is supplied valence, not autonomous discovery of importance.

## 2.4 Memory, imagination and evidence identity

Free activity carries forward to the next decision. Resetting fast activity leaves learned relationships intact, allowing a cold query to test parameter memory separately from a lingering response. Checkpoints preserve parameters, optimizer state, activity and bounded delivery bookkeeping so that restart is not confused with relearning. These contracts make retained state inspectable; task-specific retention must be measured.

Imagined alternatives settle from isolated copies of the free state. They cannot update the live state, learned parameters or witnessed-event bookkeeping. Their predictions can guide an external action, but agreement among simulated branches is not fresh evidence about the world. A branch that predicts a shock and an instrument that reads back a shock have different evidential roles. Learning from a generated target would require an explicitly different, disclosed training protocol.

This boundary is enforced by the branch operation’s lack of writes. The caller supplies observations to the learning interface; the library does not authenticate that a caller-provided target came from a physical measurement. Source identifiers authenticate neither that provenance nor the target’s truth.

An optional source identifier suppresses repeat delivery of the same observed payload while that identifier is retained, and rejects conflicting reuse of that identifier. Finite storage limits this guarantee. Distinct identifiers do not establish independent evidence; repeated reports from one sensor can share a cause. The core does not construct an authenticated causal graph, infer sensor reliability or decide whether a contradiction is noise or a changed world.

## 2.5 What an equilibrium can remember

**Fast memory and parameter memory differ.** At fixed drive and parameters, global contraction implies one equilibrium. Consequently complete settlement discards differences carried only by the initial activity. Warm starts save work but do not evade this implication. History can instead affect a persistent parameter, an explicit boundary value, or a basin choice outside global contraction. A multiple-basin system can preserve a cue under unchanged observations; learning changes its landscape and can destroy that cue while preserving the old cold input-response mapping.

**Sufficiency concerns future behaviour.** Let  $h$  be a history,  $T$  a declared deterministic update, and  $r$  its required readout. The full future signature is

$$F(h)(u) = r(T^*(h, u)), \quad U(F, i)(u) = F(i :: u), \quad r_F(F) = F(\square),$$

where  $u$  ranges over finite input continuations. Histories with identical signatures admit the same shift update and present readout. Every exact memory with state-only updates and exact readout must distinguish histories with different signatures. These statements are machine-checked <sup>3</sup>. This is a behavioural quotient, not a finite learned compression algorithm: the signature can be infinite and a callable representation may retain the whole history. A family of  $N$  behaviourally distinguishable histories requires at least  $N$  distinguishable memory states <sup>4</sup>. Finite-precision parameters therefore have a capacity limit; counting real coordinates without precision does not remove it. The required task and continuation family determine which distinctions matter. The construction does not reveal arbitrary future importance to a learner [21].

<sup>3</sup>Lean: `future_signature_one_step, every_exact_dynamic_encoding_refines_future_signature`.

<sup>4</sup>Lean: `distinguishable_family_capacity_bound`.

**A causal poset supplies dependence, not missing content.** Two records with different values can have exactly the same causal order and maximal node identifiers. The corresponding topology-only encoding fails any query that distinguishes those values <sup>5</sup>. A sufficient frontier must retain the required payloads or boundary summaries as well as causal context. Disjoint replacement writes commute under their update law; conflicting writes on one key do not <sup>6</sup>. Provenance can therefore support safe delivery and revision, but a DAG alone provides neither storage semantics, learned importance nor confluence.

**Normal form is not objective truth.** A terminating, confluent repair relation gives a unique normal form from a starting state under its hypotheses. A boundary-determined answer additionally requires all relevant history to be represented in that boundary. Neither condition implies that noisy observations were interpreted correctly. In a two-world construction, the same available reports require latent answers zero and one. Any single estimate  $z$  then satisfies

$$z^2 + (z - 1)^2 \geq \frac{1}{2},$$

so at least one squared error is at least  $1/4$  <sup>7</sup>. This is an information bound, not a stochastic noise model. Immediate perfect revision and perfect immunity to an indistinguishable first outlier cannot both be required. Acquiring another informative observation is one possible response to that uncertainty.

**Conditional basin retention.** Let  $B$  be a supplied basin, let an allowed exit from  $B$  raise a reference energy by at least  $m > 0$ , and suppose each time-varying energy  $E_t$  differs from that same reference by at most  $\epsilon$  at every state under consideration. If  $2\epsilon < m$ , an allowed step that does not increase  $E_t$  cannot leave  $B$ . Every finite accepted trace therefore stays in  $B$ , preserving any readout constant there <sup>8</sup>. A per-step bound relative only to the preceding energy is insufficient for this conclusion. The theorem assumes the exit margin and allowed moves; it does not certify the numerical neural implementation’s basins, find their margins, or prove retention under arbitrary future learning.

**Specialization needs asymmetry and useful consequences.** A deterministic permutation-equivariant update preserves an invariant initial state under invariant inputs. Exchangeable owners with identical complete states and exposures cannot differentiate by that update alone <sup>9</sup>. Random initial conditions, unequal experience or inherited asymmetry can remove this obstruction. They do not themselves establish useful division of labour. A symmetric task can have asymmetric optima. For a supplied mismatch  $r$  and connection price  $c > 0$ , minimizing  $w(r^2 + c)$  over a freely adjustable weight  $w \geq 0$  selects  $w = 0$  <sup>10</sup>. Retaining useful communication therefore needs a task constraint or consequence beyond cheaper agreement.

### 3 Current architecture and controlled evidence

The temporal experiments extend the patch interface to observed past state, private future state, action and preference ports. Their usefulness is measured by executed consequences and causal controls. Settling accuracy is reported separately from prediction, retention and control accuracy.

#### 3.1 Temporal overlap carries credit only when the constraints carry time

Let temporal patch  $t$  hold hidden state  $h_t$  and public readout  $y_t$ . A transition/readout energy is

$$E_\theta = \frac{1}{2} \sum_t (\|h_t - A\phi(h_{t-1}) - Bu_t\|^2 + \|y_t - C\phi(h_t)\|^2). \quad (1)$$

Adjacent patches overlap on the preceding state; the initial cue  $u_t$  is absent after its first event. This residual family is established temporal predictive coding [18]. Detuning the readouts toward a complete observed

<sup>5</sup>Lean: topology\_only\_is\_insufficient, maximal\_ids\_only\_are\_insufficient.

<sup>6</sup>Lean: distinct\_key\_updates\_commute, same\_key\_updates\_do\_not\_commute.

<sup>7</sup>Lean: two\_target\_squared\_errors, indistinguishable\_noise\_or\_change\_forces\_error.

<sup>8</sup>Lean: finite\_trace\_preserves\_basin, finite\_trace\_preserves\_readout.

<sup>9</sup>Lean: equivariant\_rollout\_stays\_fixed, homogeneous\_owner\_rollout.

<sup>10</sup>Lean: positive\_cost\_selects\_zero.

sequence lets discrepancy propagate across temporal overlaps, so local parameter derivatives can receive credit from later observations.

Merely allocating future patches provides no choice. For deterministic transition and readback maps and a fixed initial state, every finite defect-free path equals the forward recurrence. Lean checks this without a linearity assumption <sup>11</sup>. The statement concerns vanishing defects, not arbitrary stationary points. Useful planning therefore needs free action ports or additional constraints that make alternative consequences relevant [R83].

In the validated nonlinear sequence comparison, eight hidden coordinates and 80 learned parameters receive 1,536 updates over 16-, 32- and 64-event sequences. A distributed rotation initialization passes the 64/256-event continuation criterion in three development models, but fresh validation passes only one of three initializations on each of three-, four- and six-note motifs. A random orthogonal control passes one of nine. These are separately trained models, with every scheduled outcome retained. Each model receives 114,688 scalar teaching values in complete observed sequences, only during the detuned phases. Independent checks reconstruct derivatives, saved phase witnesses and autonomous outputs; they do not replay every training trajectory. The result supports finite temporal credit and some autonomous sequence acquisition, with neither reliable transfer nor online long-delay learning established [R83].

### 3.2 The equilibrium chooses an action using an acquired body relation

The control body obeys  $v_{t+1} = v_t + a_t + w_t$  and  $p_{t+1} = p_t + v_{t+1}$ ; the actor reads consecutive positions, not hidden velocity. Inherited coordinates  $h_t = (p_t, p_t - p_{t-1})$  expose position and displacement. Six parameters  $W = [F \ G]$  learn the next state from real position readings and executed actions. For  $E(z) = \|z - Wx\|^2/2$  with quadratic teaching, centered detuning is exactly ordinary squared-regression descent rescaled by  $1/(1 - \beta^2)$ . At  $\beta = 0.1$ , all three learned models achieve three-step held-out position MSE below  $1.3 \times 10^{-9}$  after 2,048 transitions. Matched no-history models fail. This is a deliberately transparent acquisition control, not a novel optimizer or learned coordinate system [R84].

With acquired  $F, G$  held fixed, future states and actions jointly minimize

$$E^g = \frac{q}{2} \sum_{t=0}^{H-1} \|h_{t+1} - Fh_t - Ga_t\|^2 + \frac{1}{2} \left\| h_H - \begin{pmatrix} g \\ 0 \end{pmatrix} \right\|^2 + \frac{r}{2} \sum_{t=0}^{H-1} a_t^2, \quad q = 1000, \quad r = 0.02. \quad (2)$$

The current observed boundary is fixed;  $g$  is a supplied retained preference. Adjacent three-coordinate patches exchange precision matrices and vectors. Only the first settled action is executed, after which the real position replaces the predicted readback and the actor replans. No demonstrated action sequence or external branch selector chooses the command. Soft dynamics constraints can bend imagined physics, so the evaluation measures actual executed cost and prediction defects rather than internal goal satisfaction alone.

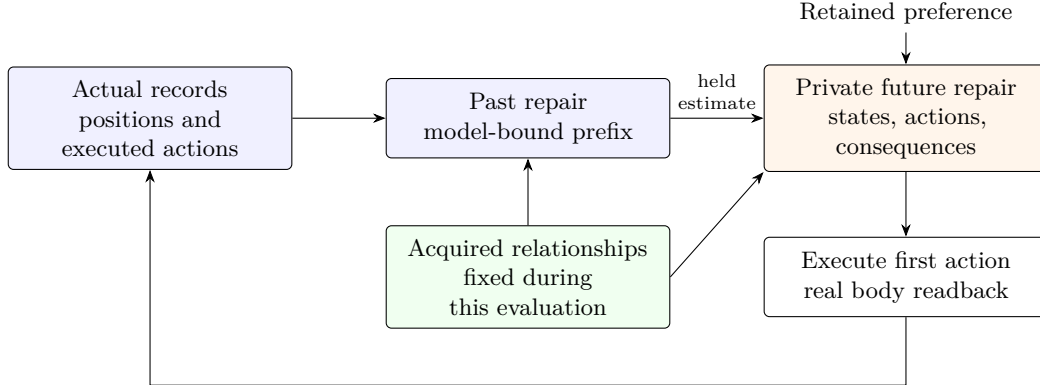


Figure 1: The tested actor’s information flow. Observed records constrain the past; a copied estimate bounds private future repair. A preference changes the proposed action without rewriting the past estimate. Real consequences close the loop. The body relations are learned from separate actual experience and then fixed for this control evaluation.

<sup>11</sup>Lean: `defectFree_unique`, `eq_forward`, `readback_eq_forward`.

| Controller with acquired body       | No disturbance | Hidden disturbance |
|-------------------------------------|----------------|--------------------|
| Three-step feedback repair          | 0.009675       | 0.020846           |
| One-step feedback repair            | 0.034174       | 0.055940           |
| Initial plan, no later feedback     | 0.009675       | 1.234675           |
| Three-step feedback, untrained body | 16.38489       | 17.78535           |

Table 2: Mean executed cost  $(p_3 - g)^2 + v_3^2 + 0.02 \sum_t a_t^2$  on the expanded control grid. Three independently acquired bodies are evaluated at  $p_0 = \pm 0.225$ ,  $v_0 = \pm 0.30$ ,  $g = \pm 0.90$  and unannounced first-action disturbance  $w_0 \in \{0, \pm 0.35\}$ . Lower is better. The disturbance column pools both signs. The retained goal, deadline and physical coordinates are supplied [R84].

All three learned models pass the declared control criteria. Masking new observations reproduces the open-loop recovery failure. The independent audit checks 1,872 episodes across the original and expanded grids and 4,680 plans, including cold restoration, goal reversal and unchanged live state during private planning. Learned feedback actions do not saturate. Forecast error after the disturbed action is  $3.01 \times 10^{-10}$ ; this excludes the unpredictable first disturbance from the forecast-accuracy measure while preserving its full observed error. These controls establish that acquired relationships, future repair and real readback each contribute to action. They do not establish autonomous goal formation or protected memory during further learning [R84].

### 3.3 Repair noisy interpretations while protecting measured records

An observation is a fixed record; an interpretation of it can be uncertain. The observer repairs latent position/displacement under actual position records and executed actions. Its Gaussian incoming message summarizes evicted history, with each measurement admitted once. The current mean and covariance are copied before the private goal/action solve. The observer has no goal argument, so a desired future cannot rewrite its factual estimate.

The tested model fixes acquired  $F, G$ , sensor variance  $R = \sigma^2$ , initial mean zero and covariance  $100I$ , and process covariance  $Q = 0.04GG^\top + 10^{-6}I$ . These are supplied model assumptions, not inferred reliability or a claim of optimality for the test disturbance distribution. Clean observations are hard position constraints, with a possibly singular posterior covariance. Under noisy readings, repairing the past improves pooled executed cost relative to raw position differences:

| Sensor standard deviation | Raw differences | Repaired past |
|---------------------------|-----------------|---------------|
| 0.02                      | 0.021238        | 0.021008      |
| 0.05                      | 0.042065        | 0.036596      |

All 1,296 episodes are retained. At the smaller noise level, cost worsens in six of 27 model/noise/disturbance strata, by at most 0.92%; at the larger level every stratum improves. Independent all-history checks recover means and covariances to  $3.91 \times 10^{-13}$  and  $1.22 \times 10^{-15}$  and reject duplicate-evidence and goal-rewrites-past mutations. This is useful conditional state estimation, not immunity to arbitrary noise or false strong observations [R85].

### 3.4 A smaller sufficient boundary under a fixed model

For a fixed linear-Gaussian model, the Gaussian prefix is a sufficient boundary for later state inference. A one-record observer is therefore compared with a three-record observer on 27 fresh 64-transition streams. The current mean, covariance and proposed three-step actions agree to  $9.05 \times 10^{-13}$ ,  $3.39 \times 10^{-14}$  and  $5.83 \times 10^{-12}$ . The observer’s persistent numeric payload, including a 32-byte model binding, falls from 272 to 208 bytes. Primary past factorizations fall from 5,265 to 1,782; additional prefix marginalizations rise from 1,701 to 1,755. Both kinds of work count. These exogenous streams test equivalence and cost, not another executed-control improvement [R85].

The compressed prefix binds to the values, shapes and dtypes of  $F, G, Q, R$  and the fixed prior. Changed models are rejected before the prefix can be silently reinterpreted. An independent sequential Gaussian oracle checks every prefix, with 378 additional dense full-history comparisons. Goal reversals preserve all past

estimates and raw records; all 432 attempted operations after model/noise changes reject without modifying the prefix. This is conditional sufficiency at fixed model coefficients. It does not solve how to revise a learned model after discarding the observations that informed it [R85].

A scalar elimination theorem makes the information boundary exact. With fixed coefficients,  $p > 0$ ,  $q \geq 0$ , and any later objective  $L$  that reads only  $y$ , define

$$E(x, y) = px^2 - 2bx + q(y - ax - u)^2 + L(y), \quad D = p + qa^2 > 0.$$

Completing the square gives

$$E(x, y) = D \left( x - \frac{b + qa(y - u)}{D} \right)^2 + p'y^2 - 2b'y + c' + L(y),$$

$$p' = \frac{pq}{D}, \quad b' = \frac{q(pu + ab)}{D}, \quad c' = \frac{qpu^2 + 2bqau - b^2}{D}.$$

The reduced expression is an attained lower bound in  $x$ . Thus two coefficients preserve later minimization over the retained boundary; the constant also preserves absolute energy. Admitting a new quadratic observation adds its precision and precision-weighted value to these coefficients. Six Lean theorems verify the identities and positivity assumptions<sup>12</sup>. The scalar proof and matrix experiment are separate evidence; neither is a theorem that an evolving nonlinear brain can compress all useful history this way [R86].

### 3.5 Retention and revision are stricter than equilibrium

A local response can be retained by constraining future parameter changes on its supporting activity. For a protected causal path, collect the presynaptic vectors for each shared map:  $\tanh h_{t-1}$  for  $A$ ,  $u_t$  for  $B$ , and  $\tanh h_t$  for  $C$ . Let the orthonormal columns of  $Q_W$  span the selected vectors for map  $W$ . The closest update to a proposed detuning step  $D_W$  that leaves these local responses unchanged is

$$D_W^{\text{protected}} = D_W(I - Q_W Q_W^T), \quad D_W^{\text{protected}} Q_W = 0.$$

This is a quadratic parameter-overlap constraint, related to established orthogonal weight modification [38]. Preserving the transition maps on the old path preserves that path by induction; preserving its readback map preserves the emitted values. Three Lean lemmas state this conditional implication without axioms<sup>13</sup>. They assume exact map agreement and do not certify the floating-point SVD, relevance selection or nearby unprotected queries.

The bases compress local response constraints, with no raw phrase or target lookup during inference. The implementation applies ordinary centered detuning and the constrained update atomically. It binds the bases to the admitted parameter history and reports consumed ranks. Protection is selected by the caller and preserves the current learned response, including its existing error. Rank exhaustion can block further learning. Importance, selective fading and justified release of conflicting constraints remain separate questions.

### 3.6 Intention-conditioned Amen phrases

A bounded real-audio task gives two streams the same eight heard bars and a single initial sign cue, followed by sixteen unobserved bars. The two intended futures are recorded phrases selected by position in pre-existing training splits from two supplied mixes. Future input is zero; targets enter only detuned learning phases. Descriptors use supplied whole-region normalization, so this is not a strict online-amplitude test. A 64-coordinate temporal network has 12,416 learned parameters. Across three initializations, all models meet the declared acquisition, cue-strength, temporal-variation, private-branch and cold-restoration checks. Mean continuation errors are 0.01423, 0.01319 and 0.01329. These are training-phrase realizations with supplied intention, not held-out-source generalization or autonomous goal formation [R88].

<sup>12</sup>Lean: `precision_pos`, `elimination_identity`, `reduced_attained`, `reduced_lower_bound`, `boundary_coefficients`, `admit_observation`.

<sup>13</sup>Lean: `protected_defectFree`, `protected_forward`, `protected_readback`.

An additional third phrase tests further learning while preserving the first response. The protected path consumes ranks 23, 8 and 24 for  $A$ ,  $B$  and  $C$ , or 28,328 numeric bytes including tolerance and parameter binding. The teaching cost uses per-output precision

$$w_j = \frac{\text{Var}_t(y_{tj}^{\text{new}}) + 0.0025}{\text{mean}_k[\text{Var}_t(y_{tk}^{\text{new}}) + 0.0025]},$$

computed from the observed new training phrase. These supplied weights add 512 checkpoint bytes; they are task weighting, not learned importance. The evaluation errors and variation gates are unweighted. After 1,536 updates on the new phrase, all three protected models pass the declared acquisition, retention, cue, variation, private-branch and cold-restoration checks. Maximum changes to the old response are  $3.92 \times 10^{-14}$ ,  $1.02 \times 10^{-14}$  and  $7.44 \times 10^{-15}$  in independent recurrence checks. New-phrase errors are 0.006226, 0.004330 and 0.003606; variation ratios are 1.464, 1.279 and 1.139, within the fixed  $[0.25, 4]$  interval. All three matched unprotected models learn the new phrase but change the old output by 0.992–1.173. Independent checks validate 36 saved phase endpoints and reject eight resealed mutations; they do not replay every intermediate update. This is conditional retention during additional phrase learning, not general lifelong memory [R89].

A separate sampler-grounded test learns actions that produce two recorded one-bar sounds. Sixteen supplied half-beat crops contain the two bars at their native 175 beats per minute. Twenty literal action coordinates specify sample choice, presence, signed offset, gain and duration. A temporal network with 64 hidden coordinates and 5,504 parameters receives a shared activation and an intention sign only at the first of eight events; later input is zero. Source-derived action sequences supply teaching targets, and known-action playback checks the sampler before training.

After 1,536 updates, all three development seeds meet the declared recall, cue, timing, restoration and private-imagination checks. All 48 sample choices across the two intentions and three seeds are correct; reversing the cue selects the other taught sequence. Independent recurrence, saved-phase and exact audio replay checks bind the learned action path to its sound. The listening files use the prescribed first seed and repeat one generated bar four times. Timing within each crop is inherited from the recording. This establishes conditional sample-sequence recall and sound realization; it does not establish original composition, auditory-feedback learning, fresh-seed generalization or retention after learning another task. A learned instrument relation, useful higher-level readback and coherent long-form development are separate requirements [R90].

The joint retention test is stricter. After 1,536 updates on a supplied third sample program, protecting both old paths preserves their outputs within  $1.71 \times 10^{-14}$  and preserves their rendered PCM exactly in the prescribed first seed. All three protected models select the eight new samples correctly, but their full action mean-square errors are 0.008584, 0.011529 and 0.011074, exceeding the declared 0.001 and quarter-baseline thresholds. Matched unprotected models fit the new program below  $6.2 \times 10^{-21}$  while changing old outputs by 0.515–0.561. None of the six models meets the joint criterion. The new program is a supplied recombination of known slices, not another external recording. The independent check covers all declared metrics, 36 saved phase replays and twelve PCM files; it does not replay every intermediate update [R91].

This limitation concerns conditioning as well as capacity. Protection consumes ranks 14, 2 and 16 for  $A$ ,  $B$  and  $C$ , or 15,432 numeric bytes, leaving 4,160 parameter directions available. The fully constrained input map fixes the cold first activity  $q$ . If  $Q$  spans the protected readout activity and  $r = (I - QQ^T)q$ , every correction with  $\Delta CQ = 0$  and  $(C + \Delta C)q = y^*$  satisfies

$$\|\Delta C\|_F \geq \frac{\|y^* - Cq\|_2}{\|r\|_2}.$$

For these three models, the residual activity norms are 0.0245–0.0322 and exact first-output correction requires readout changes of norm 21.9–28.8; the measured changes are 1.62–1.67. A rank-one correction witnesses feasibility for that first output alone. This posthoc diagnostic neither supplies a whole-program solution nor establishes rank exhaustion, impossibility or a necessary training duration [R91].

## 4 Biological abstractions and functional self-readback

Animal and human brains motivate distributed recurrent interpretation, different memory timescales, learned action consequences and specialization through unequal experience. These are functional design constraints,

not a claim that Cadence reproduces the brain’s anatomy or dynamics. Persistent-activity theories motivate carried context [9, 34]; complementary-learning-systems theories motivate separating fast experience from slower shared structure [15, 17]; predictive-coding models motivate local discrepancy signals [5, 29]. Each abstraction must earn its inclusion by retaining information or improving a measured decision.

Functional self-readback requires a causal role. Feeding a linear output  $y = Ch$  back into  $h' = Ah + Dy$  is equivalent to  $h' = (A + DC)h$ ; it does not create additional evidence or memory. The control experiment instead compares predicted consequences with new physical readings and demonstrates that masking those readings impairs recovery. A higher-level internal view would need an analogous intervention: holding observations fixed, change or remove the view and measure whether it improves the intended behavior. Stacking interfaces alone proves no such function. This paper makes no consciousness claim.

Specialization must also be useful. Exchangeable patches can receive unequal experience or break initial symmetry, but unequal weights or a named region do not establish division of labor. A learned division should improve behavior under causal lesion and transfer tests while preserving unrelated records during further learning. The fixed-model boundary and supplied-address retention results state concrete cases where information can be separated; discovering the required separation in music is a different problem.

## 5 Limits and the composition test

The most complete validated actor is a small linear control system with inherited coordinates, a supplied goal and deadline, fixed noise assumptions and a short planning horizon. Its acquired dynamics are frozen during execution. Gaussian compression is exact only relative to that fixed model. The real-audio sequence results use complete teaching sequences, three initializations and selected training phrases; held-out musical generalization is unestablished. The nonlinear temporal learner and the fixed-body actor have distinct public interfaces. Exact response protection is conditional on selected local response constraints and available plastic directions; it is not general lifelong memory under arbitrary changes.

The decisive composition test is joint: a brief intention must survive a shared heard prefix; changed intention must change the continuation; altered instrument consequences must change the next plan; imagined branches must leave witnessed state untouched; and learning new material must preserve and, when appropriate, revise old material. Context length, autonomous variation and low one-step error are separate measurements. Short intention-conditioned phrase acquisition and explicit response protection establish parts of this standard. The complete learned instrument, self-readback and long-form composition loop is unestablished. The acquired-body actor and sufficient-boundary controls identify mechanisms worth transferring, with their simplifying assumptions exposed.

Total cost includes all free and detuned phases, failed numerical attempts, matrix factorizations, communication, model acquisition, inference and any retained evidence. Local parameter derivatives do not imply low wall-clock cost. A conventional regression, filter, recurrence or controller is a necessary comparison when it solves the same stated problem. Learned developmental wiring, importance selection and robust transfer across tasks require evidence beyond the fixed small worlds measured here.

## 6 Conclusion

Overlap repair, equilibrium detuning and consequence readback can form a useful continuing actor in a controlled learned-body task. Fixed-model temporal boundaries can also compress past information exactly, with a scalar proof and independent matrix checks stating what is preserved. These are concrete instances of shared patch operations solving more than one problem. Intention-conditioned phrase acquisition and conditional protection during further learning extend these results to real musical descriptors. Automatic importance, reliable transfer, learned specialization and full intention-guided composition remain unsatisfied conditions. The architecture is judged by those joint behavioral tests, rather than by equilibrium alone.

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## A Certified recurrence and local credit

These statements concern the declared recurrent subclass; a prototype must meet their hypotheses before using their guarantees.

### A.1 Neurons, synapses, settling

A net has  $n$  neurons. Neuron  $i$  holds a potential  $v_i \in \mathbb{R}$  and publishes an activation  $s_i = \text{act}(v_i)$ . Synapses carry effective weights  $W_{ij}$  from neuron  $j$  to neuron  $i$ ; in the library the effective weight is the product of a gain, the number of contacts, a learned efficacy and a presynaptic factor, and the paper writes the product as  $W_{ij}$ . A stimulus  $c_i$  is an additive drive, with the bias absorbed into it. One settling step is

$$v_i \leftarrow (1 - dt) v_i + dt \left( \sum_j W_{ij} \text{act}(v_j) + c_i \right), \quad 0 < dt \leq 1. \quad (3)$$

The activation is a rectified, re-based logistic. With slope  $\lambda$ , threshold  $\theta$ , leak  $\ell \in [0, 1]$  and  $\sigma(x) = 1/(1 + e^{-x})$ ,

$$\rho = \sigma(-\lambda\theta), \quad r(v) = \sigma(\lambda(v - \theta)) - \rho, \quad \text{act}(v) = \frac{\max(r(v), 0)}{1 - \rho} + \ell \frac{\min(r(v), 0)}{\rho}. \quad (4)$$

Rest publishes exactly zero, the activation rises to one, and a neuron below rest publishes a small negative value so that the learning rule can move it. The learning model uses  $\lambda = 1$ ,  $\theta = 0$ ,  $\ell = 1/10$ . A settling run stops when the largest activation movement in a step falls below a tolerance or when a step cap is reached, and the library reports the residual of equation (3) at the final state. Adaptation, a slow variable subtracted from the drive, produces rhythmic dynamics; the theorems below are stated without it.

A declared neural graph can include sensory, internal and motor coordinates. Joint settlement can complete a partial reading when its learned constraints support that completion; region names alone assign no function.

**Intuition: each neuron checks its own account.** Imagine an arm whose visual input has moved while its motor and body predictions describe the preceding position. Each neuron compares its held potential with the drive its neighbours and senses supply. Equation (3) moves the potential partway toward that drive; the published activation changes what neighbouring neurons receive, and they make the same comparison. The resulting computation is a feedback process with a measurable discrepancy at every step. In the observer-patch description, each bounded patch owns state, communicates through declared ports, reads back its consequences and retains records. A settling certificate and the public evidence bundle make that self-reading process checkable. The equilibrium reconciles the equations the model implements; whether those equations describe the arm well is a separate empirical question.

## A.2 From observable events to a global normal form

OPH distinguishes internal descriptions from shared readings [24–26]. A visual patch may locate an object in camera coordinates and a motor patch in body coordinates. Agreement concerns the position after translation to a shared interface. A change of internal coordinates and corresponding interface map is a gauge change when it preserves every declared public reading. Gauge-invariant repair corrects disagreement in those readings.

For local states  $x_i$  and readout maps  $\pi_{i,e}$  into a common interface on an overlap  $e = \{i, j\}$ , agreement means

$$\pi_{i,e}(x_i) = \pi_{j,e}(x_j).$$

A repair preserves the observations held fixed. OPH’s local-repair criterion requires every allowed fixed exterior to admit a consistent interior. A *normal form* has no repair left to make. For a terminating discrete relation on the observable quotient, local confluence gives one normal form from each starting state. Completeness, meaning that normal states coincide with consistent states, makes it a global solution. Local compatibility alone need not imply global consistency. For different interiors with the same protected observations to yield the same answer, the compatible consistent states must also form one equivalence class [25, 26].

**What counts as an observable event?** For a finite classical record, events are sets of possible readings, closed under union, intersection and complement. OPH’s verified projection-event calculus supplies conditional expectations that discard internal distinctions while preserving the probabilities of the declared partition events [22]. Cadence uses classical activation ports and writable records; the quantum projection and conditioning machinery is mathematical background, without a claimed implementation in this neural model.

**The neural specialization.** Hold the weights fixed, protect  $B(c, v) = c$ , and define consistency by  $v = W \text{ act}(v) + c$ . Under Cadence’s contraction condition, every initial state at a fixed input approaches the same equilibrium. The residual and stimulus bounds quantify accuracy and sensitivity. This realizes an observation-determined equilibrium: local equations determine one global answer with a checkable error bound. Numerical convergence replaces finite exact termination; the synchronous proof does not establish arbitrary asynchronous or transactional repair guarantees. Learning determines whether the answer is useful.

## A.3 The certificate: how much work a decision needs

Write  $F_c(v)$  for the right-hand side of (3) at stimulus  $c$ ,  $\|\cdot\|_\infty$  for the sup norm on  $\mathbb{R}^n$ ,  $\rho_W = \max_i \sum_j |W_{ij}|$  for the largest absolute incoming weight sum (the row mass), and  $L$  for a Lipschitz constant of  $\text{act}$ .

**Theorem A.1** (settling certificate). *If  $0 < dt \leq 1$  and  $L\rho_W < 1$ , then  $F_c$  is a contraction of  $(\mathbb{R}^n, \|\cdot\|_\infty)$  with constant  $q = 1 - dt(1 - L\rho_W) < 1$ .*<sup>14</sup>

**Corollary A.2** (unique equilibrium). *Under the hypotheses of Theorem A.1 the net has exactly one equilibrium  $v_c^*$ , and every fixed point of  $F_c$  equals it; with zero stimulus and  $\text{act}(0) = 0$  that equilibrium is rest.*<sup>15</sup>

<sup>14</sup>Lean: `settle_contracting`.

<sup>15</sup>Lean: `equilibrium_unique, silence`.

**Theorem A.3** (a priori and a posteriori certificates). *Under the hypotheses of Theorem A.1, for every start  $v$  and every  $k$ ,*

$$\|F_c^k(v) - v_c^*\|_\infty \leq \frac{q^k}{1-q} \|v - F_c(v)\|_\infty, \quad \|F_c^k(v) - v_c^*\|_\infty \leq \frac{\|F_c^k(v) - F_c^{k+1}(v)\|_\infty}{1-q}.$$

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A residual below  $\epsilon$  at the returned state certifies a potential error of at most  $\epsilon/(1-q)$ , and an activation error of at most  $L$  times that. This is an error bound relative to the model's equilibrium, not a certificate of prediction accuracy. Banach's theorem is the whole proof [4]; the activation-specific condition is given in Theorem A.6. A numerical control checks the bound on 57,600 cases [R8].

**Theorem A.4** (the equilibrium is Lipschitz in the stimulus, and warm starts pay). *Let  $F_c$  and  $F_{c'}$  satisfy Theorem A.1 with the same constant  $q$  and let  $\delta = \|c - c'\|_\infty$ . Then*

$$\|v_c^* - v_{c'}^*\|_\infty \leq \frac{dt \delta}{1-q}, \quad \|F_{c'}^k(v_c^*) - v_{c'}^*\|_\infty \leq \frac{dt \delta q^k}{1-q} \quad \text{for every } k.$$

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This is the theorem behind adaptive work. Starting from the old equilibrium, the number of steps that reaches accuracy  $\epsilon$  after an input change  $\delta$  is at most the nonnegative ceiling of  $\log(dt \delta / ((1-q)\epsilon)) / \log(1/q)$ : logarithmic in the size of the change, zero when the change is zero. The bound is sufficient rather than a prediction of semantic difficulty, and checking a returned state has its own cost.

**Intuition: reuse the answer, then bound the repair.** Contraction says that two possible internal states move closer under the same input: after one step their largest difference is at most  $q$  times its previous value. With  $q = 1/2$ , every step halves that uncertainty. If the residual at a returned state is  $10^{-3}$ , the remaining potential error is at most  $2 \cdot 10^{-3}$ ; the factor  $1/(1-q)$  sums all the smaller corrections that can follow. When a sensor changes slightly, the preceding equilibrium is a useful starting point because Theorem A.4 bounds how far the required answer can move. This can save repeated work on a slowly changing stream. It gives little advantage after a large change, and convergence can be expensive when  $q$  is close to one. A unique equilibrium also erases differences in starting state: lasting history must enter through records, traces or a changed drive, rather than through the warm start alone.

**Theorem A.5** (bounded damage). *Let  $|\text{act}| \leq A$ , let  $S$  be a set of neurons, let  $W^S$  be  $W$  with the columns in  $S$  set to zero, and let  $\mu_S = \max_i \sum_{j \in S} |W_{ij}|$ . If  $F_c$  satisfies Theorem A.1, then the lesioned map satisfies it with the same constant, and  $\|v^* - v^{*,S}\|_\infty \leq dt \mu_S A / (1-q)$ .<sup>18</sup>*

**Theorem A.6** (the library's neuron). *For  $\lambda \geq 0$  and  $\ell \geq 0$ , the activation (4) is Lipschitz with constant  $\frac{\lambda}{4} \max(\frac{1}{1-\rho}, \frac{\ell}{\rho})$ , satisfies  $|\text{act}| \leq \max(1, \ell)$ , and publishes exactly zero at rest. Under the learning model the constant is  $1/2$  and the bound is 1, so every net whose row mass is below 2 settles by a certified contraction for every  $0 < dt \leq 1$ .<sup>19</sup>*

For the hypothesis, the certificate makes the numerical state of a changing brain checkable: when its conditions hold, a returned decision carries a bound on its distance from equilibrium. This bound certifies the solve. It does not certify the accuracy of the learned body model or the safety of an action.

## A.4 Learning by two equilibria

The free phase settles under the stimulus alone to  $s^0$ . The positive nudged phase settles from  $s^0$  with an extra drive of strength  $\beta$  on the output neurons toward a target: for a quadratic nudge the drive is  $\beta(\text{target}_i - s_i)$ ,

<sup>16</sup>Lean: `apriori_bound`, `aposteriori_bound`.

<sup>17</sup>Lean: `equilibrium_lipschitz_stimulus`, `warm_start_bound`.

<sup>18</sup>Lean: `lesion_bound`, `lesion_contracting`.

<sup>19</sup>Lean: `activation_lipschitz`, `activation_abs_le`, `activation_zero`, `learningActivation_lipschitz`, `learning_model_certified`.

for a cross-entropy nudge  $\beta(\text{target}_i - p_i)$  with  $p$  the softmax of the outputs at temperature  $T$ , one softmax per declared slot. The centred estimator settles a negative nudged phase at  $-\beta$  from the same free state. With  $s^+$  and  $s^-$  the two nudged equilibria, the contrast at synapse  $(i, j)$  and at neuron  $i$  is

$$\Delta_{ij} = \frac{1}{B} \sum_b \frac{s_{b,i}^+ s_{b,j}^+ - s_{b,i}^- s_{b,j}^-}{2\beta}, \quad \Delta_i = \frac{1}{B} \sum_b \frac{s_{b,i}^+ - s_{b,i}^-}{2\beta}. \quad (5)$$

Reciprocal synapses share one parameter and receive the mean of their two contrasts; a cap bounds every efficacy.

**Lemma A.7** (locality). *There is a fixed function  $g$  of four real numbers such that the contrast of synapse  $(i, j)$  equals  $g(s_i^+, s_j^+, s_i^-, s_j^-)$ ; the contrast is symmetric in  $i$  and  $j$ ; and if the two compared equilibria agree, every contrast is zero.*<sup>20</sup>

The lemma describes the information used by the update. Reverse-mode differentiation retains or re-computes intermediate activations; the equilibrium contrast instead reads each synapse’s endpoints in the perturbed states. This can reduce storage when differentiating through many settling iterations. Both perturbed equilibria must be computed, and implicit differentiation offers another way to avoid retaining the iteration history [3].

**Fact A.8** (the contrast is the gradient). *On a smooth stable equilibrium branch with symmetric effective recurrent weights, converged phases and no adaptation, the limit  $\beta \rightarrow 0$  of the contrast is minus the gradient of the nudge’s loss with respect to the weight of a reciprocal pair [30]; the centred estimator cancels the first-order bias in  $\beta$  [16]. The library’s tests check the identity numerically: at  $\beta = 10^{-4}$  and phase residuals below  $10^{-11}$  the contrast equals the finite-difference gradient to a relative tolerance of  $10^{-5}$ .*

Free and nudged phases are the deterministic form of the two phases of the Boltzmann machine [1, 19], equivalent to backpropagation in a layered net [36], and the local-activation-difference rule of generalized recirculation [27]. Reward uses the three-factor form: an eligibility trace  $e \leftarrow \gamma \lambda e + \Delta$  of the same contrast, a linear critic with its own trace, and one broadcast error  $\delta = r + \gamma V(s') - V(s)$ , with the update  $\eta \delta e$  [8, 32, 33].

**Intuition: let a small correction travel through the circuit.** The target gently pushes the outputs, and recurrence transmits that change to the rest of the net. A synapse compares how strongly its two endpoints coactivate under the positive and negative pushes. Their difference tells it which direction the connection should move under the assumptions of Fact A.8. Credit reaches the synapse through the circuit’s changed state, so the update itself needs only its endpoints. There is a cost: the learner must solve the perturbed equilibria, and a locally computed update can be slower than a conventional backward pass. An eligibility trace carries that local comparison forward when reward arrives later; it does not guarantee that one delayed scalar identifies the right cause. The advantage is the form and storage of the update, with learning quality established by the controlled comparisons.

## B Optional record algebra and capacity

A record cortex keeps, for each of its cells, what followed the readings that cell took part in. A reading  $x \in \mathbb{R}^d$  of declared source neurons is made mean-free by a running mean  $\bar{x}$ , projected by a fixed random matrix  $R$  onto  $D \gg d$  cells, rectified, and reduced to the  $a$  strongest entries at unit length,

$$k(x) = \text{kwta}_a(\max(R(x - \bar{x}), 0)). \quad (6)$$

Each predicted field holds one table  $M \in \mathbb{R}^{D \times m}$ . The prediction is the read  $k(x)M$ , and the witnessed outcome  $y$  is written by the delta rule [12, 35] at rate  $\eta$ ,

$$M \leftarrow M + \eta k(x)^\top (y - k(x)M). \quad (7)$$

Consequence fields use a slow rate so that a regularity is averaged over many outcomes; valued fields such as reward use rate one, so that one exposure writes. The optional record store is distinct from the neural core and from the fixed-model Gaussian boundary.

<sup>20</sup>Lean: `contrast_local`, `contrast_symm`, `contrast_eq_zero_of_agree`.

**Theorem B.1** (one-shot memory and its interference law). *For any record  $M$ , keys  $k$ ,  $q$ , value  $y$  and rate  $\eta$  over a commutative ring: (i) the read at  $q$  after the write changes by  $\eta(q \cdot k)$  times the correction  $y - kM$ ; (ii) if  $k \cdot k = 1$ , the read at  $k$  after a write at rate one is exactly  $y$ ; (iii) if  $q \cdot k = 0$ , the read at  $q$  is unchanged; (iv) if  $k \cdot k = 1$ , writing the same correct observation twice at rate one changes nothing.*<sup>21</sup>

**Theorem B.2** (how far one write reaches). *Over the reals, the change a write at  $k$  makes to the read at  $q$  has sup norm exactly  $|\eta| |q \cdot k|$  times the sup norm of the correction, and an overlap of at most  $\epsilon$  bounds it by  $|\eta| \epsilon$  times that norm.*<sup>22</sup>

**Theorem B.3** (exact storage and capacity). *After writing  $N$  pairwise orthogonal unit keys in sequence at rate one, every one of them reads its own value, whatever the initial record was; and over a field at most  $d$  keys can be pairwise orthogonal unit vectors in a  $d$ -dimensional key space.*<sup>23</sup>

**Theorem B.4** (pattern separation). *Two keys whose supports are disjoint are orthogonal, so a write at one leaves the read at the other exactly unchanged; a family of unit keys with pairwise disjoint supports, written in sequence at rate one, is stored exactly from any initial record; and a code of width  $D$  with  $a$  active cells admits at most  $\lfloor D/a \rfloor$  such keys.*<sup>24</sup>

**Intuition: a correction follows the overlap of two addresses.** Suppose a reading activates two cells equally, giving a unit key  $k = (1, 1, 0, 0)/\sqrt{2}$ . A write at rate one corrects that reading in a single step. A second reading with key  $q = (0, 0, 1, 1)/\sqrt{2}$  shares no cells and keeps its prediction exactly. A third,  $q = (0, 1, 1, 0)/\sqrt{2}$ , shares one cell and receives half the first reading’s correction, since  $q \cdot k = 1/2$ . This is the interference law in a small example: the same overlap that permits generalisation can overwrite another fact.

The four theorems explain why a useful code can learn quickly from a stream: each observation directly corrects the relevant prediction while disturbing other predictions in proportion to their key overlap. Sparsity alone guarantees neither distinct addresses nor retention; two sparse keys can be identical, and finite capacity limits exact separation. With an appropriate code, direct writes can need fewer observations than fitting the same associations in distributed weights. The record theorem does not by itself prove a replay requirement for another architecture. The source-bound application receipts document measured uses of these records [R51,R52,R53,R54].

**Theorem B.5** (attention is one read of a record). *Let  $M = \sum_t k_t^\top y_t$  be the additive record of  $N$  key/value pairs. The read at query  $q$  is  $qM = \sum_t (q \cdot k_t) y_t$ , the stored values weighted by the query/key dot products. If the keys are one-hot at distinct addresses, the read at a one-hot query is exactly the stored value, for every  $N$  up to the size of the address space.*<sup>25</sup>

The first sentence is unnormalised linear attention [2, 11, 31]; the softmax read of a transformer differs by an exponential kernel and a normaliser, and modern Hopfield networks show that it is one step of an energy network [13, 14, 28]. The residual write of equation (7) is the delta rule over a sequence [37]. The identity covers additive dot-product attention, not the general softmax operation. Unlike an additive write, the residual write corrects its current prediction.

## C Wiring and a declared cost model

**Theorem C.1** (the certificate under evolution). *Let  $W$  satisfy Theorem A.1 with row mass  $\rho$ . (i) Every  $W'$  with  $|W'_{ij}| \leq |W_{ij}|$  for all  $i, j$  satisfies it with the same constant: pruning a synapse, masking a projection, lowering a gain and scaling by any  $|g| \leq 1$  are instances. (ii) If  $V$  has row mass  $\mu$  and  $L(\rho + \mu) < 1$ , then  $W + V$  settles by a certified contraction with constant  $1 - \text{dt}(1 - L(\rho + \mu))$ : growth by new projections stays certified inside the row-mass budget. (iii) Every member of a population whose genomes share one row-mass bound settles by one certificate, whatever its individual wiring.*<sup>26</sup>

<sup>21</sup>Lean: read\_write, read\_write\_self, read\_write\_orth, repeat\_write\_stable.

<sup>22</sup>Lean: read\_write\_dist, read\_write\_dist\_le.

<sup>23</sup>Lean: read\_writeUpTo, capacity\_le.

<sup>24</sup>Lean: dotProduct\_eq\_zero\_of\_disjoint\_support, read\_write\_of\_disjoint, read\_writeUpTo\_of\_disjoint, disjoint\_family\_card\_le.

<sup>25</sup>Lean: read\_hebb, one\_hot\_recall.

<sup>26</sup>Lean: rowMassLE\_of\_abs\_le, rowMassLE\_smul, rowMassLE\_add, mutation\_contracting, growth\_contracting, population\_contracting.

The row-mass budget certifies convergence of admitted recurrent weights. It does not establish useful specialization or certify a changing nonlinear memory basin.

## C.1 Cost

Fix the declared cost model. A patch net with  $E$  synapses that settles in  $k$  steps costs  $kE$  multiply-adds per decision, plus  $Dm$  for a record read and the same for a write, and holds  $n$  potentials and  $Dm$  numbers whatever has happened before. A transformer with  $\ell$  layers of width  $d$  over a window of  $T$  tokens costs about  $\ell(12d^2 + 2Td)$  multiply-adds per token and holds a key/value cache of  $2\ell Td$  numbers.

**Theorem C.2** (what grows and what does not). *Under that model, the arithmetic a transformer spends on a stream of  $T$  tokens is  $\ell(12d^2T + dT(T - 1))$ , which grows quadratically in  $T$ , and its cache grows without bound. The arithmetic a patch net spends is  $T(kE + 2Dm)$  and its state is constant in  $T$ . For every choice of positive  $\ell, d, k, E, D, m$  there is a horizon beyond which the patch net’s total is the smaller of the two.*<sup>27</sup>

The theorem is arithmetic about the declared model, not a claim that the two objects compute the same function. It states where the shapes differ: one of them re-reads its past, the other has written its past into a fixed store. Measured costs depend on the numerical solver, query accuracy and full controller.

**Intuition: compression trades history for a bounded state.** For a repeated body-control task, a learner may need the present position and a learned map from actions to consequences, rather than every observation that established the map. Records and traces carry that summary between decisions. Their fixed size explains the linear stream cost in the theorem, and also imposes a limit: they cannot preserve arbitrarily many independent facts exactly. The quadratic comparison assumes attention to an expanding history; a transformer with a fixed window also has linear total cost in stream length. Whether retaining the summary is better depends on what the task requires, the cost of finding the representation, the achieved error and the measured latency.

## D Theorem map and the Lean library

Every Theorem, Lemma and Corollary in the paper names the Lean declaration that proves it. The public Cadence library includes the certified recurrence, activation, local contrast and wiring results, together with the representation, noise, basin, specialization and temporal-overlap boundaries [20]. The optional linear-record algebra has a separate import; it is not a component of **PatchNet**. These sources pin Lean 4.29.1 and Mathlib at the matching release. Their axiom audit allows only propositional extensionality, choice and quotient soundness, with no admissions or project-specific axioms. The historical cost results are retained in the full manuscript’s proof collection; exploratory structural-memory, readable-trace, causal-path and quadratic-boundary results have separate sources and verification in the accompanying evidence [21]. The causal-path, quadratic-boundary and protected-path lemmas are included in the public package’s 169 audited statements; other exploratory collections remain separate. The full manuscript build checks these additional sources as well. None of these collections is a formal refinement proof of the floating-point experiments or the complete library interface.

Table 3: The theorem map. Settling theorems hold for Lipschitz activations; the Activation module instantiates them for this neuron. Memory identities hold over commutative rings, the capacity bound over fields, the cost statements over the naturals.

| paper   | module and declaration                                                | content                                                                 |
|---------|-----------------------------------------------------------------------|-------------------------------------------------------------------------|
| Thm A.1 | Settling, <code>settle_contracting</code>                             | contraction with constant $1 - dt(1 - L\rho_W)$                         |
| Cor A.2 | Settling, <code>equilibrium_unique</code> , <code>silence</code>      | any fixed point is the equilibrium; rest is the equilibrium of no input |
| Thm A.3 | Settling, <code>apriori_bound</code> , <code>aposteriori_bound</code> | the two error certificates                                              |

<sup>27</sup>Lean: `attention_total`, `attention_total_ge_quadratic`, `record_total`, `record_state_const`, `exists_crossover`.

| paper   | module and declaration                                                                                                       | content                                                                                                                                                          |
|---------|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Thm A.4 | Settling, equilibrium_lipschitz_stimulus, warm_start_bound                                                                   | the equilibrium moves by at most $dt \delta / (1 - q)$ ; $k$ warm steps leave $dt \delta q^k / (1 - q)$                                                          |
| Thm A.5 | Settling, lesion_bound, lesion_contracting                                                                                   | a lesion moves the equilibrium by at most $dt \mu_S A / (1 - q)$                                                                                                 |
| Thm A.6 | Activation, activation_lipschitz, activation_abs_le, activation_zero, learningActivation_lipschitz, learning_model_certified | the neuron’s Lipschitz constant, range and rest; row mass below two certifies the learning model                                                                 |
| Lem A.7 | Credit, contrast_local, contrast_symm, contrast_eq_zero_of_agree                                                             | locality, symmetry, zero contrast for identical states                                                                                                           |
| Thm B.1 | Memory, read_write, read_write_self, read_write_orth, repeat_write_stable                                                    | interference law, one-shot write, orthogonal invariance, idempotence                                                                                             |
| Thm B.2 | Sparse, read_write_dist, read_write_dist_le                                                                                  | the exact size of one write’s effect on another read                                                                                                             |
| Thm B.3 | Memory, read_writeUpTo, capacity_le                                                                                          | exact storage of an orthonormal family; at most $d$ such keys                                                                                                    |
| Thm B.4 | Sparse, dotProduct_eq_zero_of_disjoint_support, read_write_of_disjoint, read_writeUpTo_of_disjoint, disjoint_family_card_le  | disjoint supports never interfere; a disjoint family is stored exactly; a code of width $D$ with $k$ winners holds at most $\lfloor D/k \rfloor$ of them         |
| Thm B.5 | Memory, read_hebb, one_hot_recall                                                                                            | the additive read is linear attention; one-hot recall is exact                                                                                                   |
| Thm C.1 | Wiring, rowMassLE_of_abs_le, rowMassLE_smul, rowMassLE_add, mutation_contracting, growth_contracting, population_contracting | pruning, masking and rescaling keep the certificate with the same constant; growth keeps it inside the row-mass budget; one bound certifies a whole population   |
| Thm C.2 | Cost, attention_total, attention_total_ge_quadratic, record_total, record_state_const, exists_crossover                      | the transformer’s arithmetic over a stream is quadratic in its length and its cache unbounded; the patch net’s is linear with constant state; a crossover exists |

Three further results are used as facts and are Lean-checked in the observer-patch corpus rather than in this package [23, 25]: local repair to consensus on a federation of patches computes every finite Boolean function (`echosahedral_universality`); a work-conserving scheduler on a ranked repair system terminates without a fairness premise; and the accepted repair steps of a fixed input-independent federation number at most  $n(n + 1)/2$ , order-sharp. No embedding of those discrete programs into this neuron model is supplied.

## E Evidence map

Empirical numbers are read from the named receipts copied into the evidence package, with each receipt’s own digest where it has one, its implementation version or source hashes, and the sha256 of the copied bytes recorded in the package manifest. Quoted external results have bibliographic citations. Exact mathematical constants follow their stated definitions and hypotheses. The memory-proof sources are copied under the manifest label `mission-memory-proofs`.

Table 4: Receipt identifiers and manifest labels. The library column names the release whose sources the receipt binds; OPN identifies the earlier observer-patch-nets library. The manifest preserves exact source references and copied-file hashes.

| id | manifest label     | receipt                                                      | library |
|----|--------------------|--------------------------------------------------------------|---------|
| R1 | recall             | associative recall over 128 pairs, two transformer arms      | 0.5.0   |
| R2 | revision-memory    | memory under a revision stream, five arms, three seeds       | 0.8.1   |
| R3 | pattern-separation | correlated keys under revision streams, six arms, five seeds | 0.9.0   |

| id  | manifest label                | receipt                                                                                                                         | library      |
|-----|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------|
| R5  | digit-span                    | digit span, six arms, fade sweep                                                                                                | 0.7.0        |
| R6  | n-back                        | <i>n</i> -back, four arms, two baselines                                                                                        | 0.7.0        |
| R8  | adaptive-depth                | compute follows change: warm steps, budget, bounds, three row masses, five seeds                                                | 0.9.0        |
| R10 | event-repair                  | event-driven repair on the fly graph, 135 conditions                                                                            | 0.8.1        |
| R12 | surprise                      | settling steps against entropy                                                                                                  | 0.5.0        |
| R13 | matched-learning              | three learning rules on one 48-neuron mechanism                                                                                 | 0.8.1        |
| R16 | lesion                        | damage and recovery, three seeds                                                                                                | 0.5.0        |
| R17 | permuted-mnist                | thirty permuted tasks, six arms                                                                                                 | 0.5.0        |
| R21 | worm-learns                   | the worm learns an odour, three seeds, lesions, swap                                                                            | 0.8.1        |
| R22 | fly-counterfactuals           | whole-fly counterfactual screening                                                                                              | 0.8.1        |
| R23 | opn                           | fly wiring against shuffled, conformance                                                                                        | OPN          |
| R24 | mouse                         | mouse labyrinth, connectome against shuffled and a network                                                                      | 0.7          |
| R26 | player                        | self-play of the many-games brain                                                                                               | 0.7          |
| R27 | player                        | sixteen games at once with a dopamine floor                                                                                     | 0.8          |
| R30 | ladder-v0.5.0                 | digits, MNIST, Connect Four, Pong, text, sign, chorales, cart-pole, worm                                                        | 0.4          |
| R31 | sizing                        | text width sweep, per-run files                                                                                                 | 0.4 to 0.7   |
| R32 | author                        | language pilots at level one                                                                                                    | 0.7          |
| R33 | hopper, pong                  | control gates, logs                                                                                                             | 0.5.0        |
| R34 | repeat-previous, slippery-ant | partially observed cards; slippery ant                                                                                          | 0.7 to 0.8.1 |
| R36 | split-mnist                   | split MNIST, one head                                                                                                           | 0.5.0        |
| R37 | worm-probe                    | supplied worm mechanism under lesions                                                                                           | 0.8.1        |
| R41 | ep-inputs                     | fixed-input derivatives and free/free asymmetry control                                                                         | 0.9.0        |
| R42 | rehearsal                     | split and permuted MNIST at disclosed memory and work budgets                                                                   | 0.9.0        |
| R43 | content-prototypes            | noisy cues, fixed prototypes, aliases and capacity                                                                              | 0.9.0        |
| R44 | policy-credit                 | delayed choices, loss-policy correction and score control                                                                       | 0.9.0        |
| R45 | sequence-readback             | causal held-out text and validation grids                                                                                       | 0.9.0        |
| R46 | drift-diagnostic              | thirty tasks and equal-budget reset forks                                                                                       | 0.9.0        |
| R51 | experience-arm                | the arm: five acceptance seeds, twelve gates, six controls                                                                      | 0.9.0        |
| R52 | experience-world              | the world: five acceptance seeds, eleven gates, eight controls                                                                  | 0.9.0        |
| R53 | experience-connect-four       | Connect Four: five acceptance seeds, twelve gates, the threat read and the memory of proved positions                           | 0.9.0        |
| R54 | experience-artist             | the artist: five acceptance seeds, thirteen gates, eight controls                                                               | 0.9.0        |
| R55 | experience-comparisons        | the same lives with the world model replaced by a multilayer network or a transformer, with and without a replay ring           | 0.9.0        |
| R56 | solver-bench                  | Connect Four against graded solver opponents, every move scored by a perfect solver                                             | 0.9.0        |
| R57 | instrument                    | the instrument: five acceptance seeds, ten gates                                                                                | 0.9.0        |
| R58 | composer-stage                | the chip composer: five acceptance seeds, thirteen gated and seven open measures                                                | 0.9.0        |
| R60 | connect-four-bench            | Connect Four against a published AlphaZero recipe, the threat-read ablation, the trap suite                                     | 0.9.0        |
| R61 | wiring-search                 | wiring search on four families and on the composer's streams: evolution, random search, hand-found wirings, intervention states | 0.9.0        |
| R62 | scaling-curves                | fourteen arms on four task families at four budgets, preregistered floors, retention read-back                                  | 0.9.0        |
| R63 | combined-learner              | banks beside a gradient patch with a guard, ten arms, four budgets                                                              | 0.9.0        |
| R64 | evolved-development           | the evolution loop over wiring and development policy: contract and calibration                                                 | 0.9.0        |
| R65 | composer-curves               | the chip composer as curves over a life: three wirings, two sizes, two life lengths, four seeds                                 | 0.9.0        |
| R66 | player-curves                 | the 1942 records controller and a PPO convolutional net at matched frames, three seeds                                          | 0.9.0        |
| R67 | world-evolution               | the evolution world: five sweeps over rules and prices, two seeds each, censuses and lifetimes                                  | 0.9.0        |
| R68 | maestro                       | the 68-million-synapse musician: receipt, model card and evidence sheet with the recurrent control                              | 0.9.0        |
| R69 | player-1943                   | the 1943 imitation player and its practice on 1942, frozen tests on eight starts                                                | 0.9.0        |
| R70 | worm-contact                  | pinned browser sources, complete boundary-contact trace and matched touch-off control                                           | 0.8.1        |

| id  | manifest label              | receipt                                                                                                                                               | library     |
|-----|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| R71 | mission-basin               | nonlinear cue persistence, opposing evidence and continued-learning stress; source, full receipt and independent verification                         | OPN         |
| R72 | mission-specialization      | nine-patch completion and warm revision; source, full receipt and independent verification                                                            | diagnostic  |
| R73 | mission-structural          | normalized-contact repair, continuous controls and shared-connection failure; full receipt and verification                                           | diagnostic  |
| R74 | mission-active-evidence     | exact finite selective-inspection reference; frozen model, exhaustive enumerator and receipt                                                          | reference   |
| R75 | library-memory              | reciprocal library core without temporal boundary; three-seed acquisition, retention, cue and continuation checks with archived sources               | core        |
| R76 | library-temporal-boundary   | same declared library tasks with held temporal overlap; finite context, cold-acquisition failures and archived sources                                | core        |
| R77 | amen-patch-features         | fixed-seed recorded-audio feature prediction with and without temporal overlap; matched autoregressive controls, arrays and source snapshots          | core        |
| R78 | library-selective-retention | absent-boundary, fixed-input and calibrated selective-plasticity tests; complete outcomes, checkpoints, sources and independent endpoint verification | core        |
| R79 | scalar-continuation         | periodic continuation with current, temporal, explicit-history and supplied-clock inputs; frozen protocol, full receipt and arithmetic verification   | core        |
| R80 | readable-trace              | six scalar trace-repair lemmas; pinned sources and compiled axiom audit                                                                               | formal      |
| R81 | intent-control              | brief-cue motor/consequence test, matched trace ablation, private branches, continued-state acquisition and linear-decoder diagnostics                | core        |
| R82 | intent-episodic             | independent episodic intentions with fresh-state training; trace ablation and supplied intention/position acquisition control                         | core        |
| R83 | temporal-repair             | causal-path uniqueness proof; linear and nonlinear temporal credit, initialization comparison and fresh motif validation                              | diagnostic  |
| R84 | temporal-actor              | linear body acquisition by centered detuning; joint action planning, actual feedback, cold restoration and independent execution audits               | diagnostic  |
| R85 | temporal-observer           | noisy past-state repair and one-record Gaussian-prefix equivalence; actual records, model binding, controls and independent all-history audits        | diagnostic  |
| R86 | quadratic-boundary          | six scalar fixed-model quadratic elimination and observation-admission theorems; compiled axiom audit                                                 | formal      |
| R87 | revision-exposure           | fresh crossed static/warm exposure and gradient/Newton repair; complete memory outcomes and independent endpoint arithmetic                           | diagnostic  |
| R88 | amen-temporal-capacity      | real-audio intention-conditioned finite paths; all-seed acquisition, retention controls and independent endpoint/recurrence checks                    | capacity    |
| R89 | amen-temporal-retention     | local response constraints, new-phrase learning, protected-path Lean induction, sample listening and independent checks                               | conditional |
| R90 | amen-sample-events          | cue-conditioned recall of two source-derived sample sequences; known-action controls, source custody and independent neural/audio replay              | capacity    |
| R91 | amen-sample-retention       | matched further learning of sampled actions, exact old-response protection, first-event conditioning and independent neural/PCM checks                | bound       |

## F Notation

$n$  neurons;  $v_i$  potential;  $s_i = \text{act}(v_i)$  activation;  $W_{ij}$  effective weight from  $j$  to  $i$ ;  $c_i$  stimulus with bias absorbed;  $dt$  step;  $\rho_W$  largest absolute incoming weight sum;  $L$  Lipschitz constant of  $\text{act}$ ;  $q$  contraction constant;  $v^*$  equilibrium;  $\beta$  nudge strength;  $s^\pm$  nudged equilibria;  $\Delta$  contrast;  $\eta$  learning rate;  $M$  record table;  $k, q$  keys;  $y$  value;  $D$  code width;  $a$  active cells of a code;  $k(x)$  the code of a reading, which is its key;  $R$  the expansion;  $e$  eligibility trace;  $\delta$  the broadcast reward error, or the size of a stimulus change where stated;  $\gamma, \lambda$  discount and trace decay;  $\mu$  the row mass of added projections;  $\ell, d, T$  layers, width and window of the transformer in the cost model.